PRODUCT REQUIREMENTS DOCUMENT

IT Asset & Inventory Management System

Version: 1.0
Document status: Initial baseline for review and development planning
Application type: Internal web application
Operating model: Single-company deployment
Intended audience: Project owner, business stakeholders, UI/UX, frontend and backend developers, QA

Document purpose
Define the product requirements, users, workflows, business rules, scope, technical direction and acceptance expectations. This document will be used as the source of truth for preparing the development plan and implementing the application.

Prepared as a practical project baseline. Detailed implementation choices will be confirmed during technical planning without changing the approved business requirements.

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1. Project Overview

The IT Asset & Inventory Management System is an internal web application for a single organization. It provides a centralized place to record IT equipment, monitor stock, manage vendors and purchases, assign equipment to employees, track returns and manage maintenance.

The system covers the lifecycle of an IT asset from purchase and receipt through availability, assignment, return, maintenance and retirement. It must also support quantity-based tracking for accessories or consumable items that are not individually serialized.

1.1 Product vision

Give the organization a reliable and easy-to-use view of what IT equipment it owns, where the equipment is, who is using it, what is available and what needs attention.

1.2 Product context

Item	Definition
Deployment	One organization using one shared application instance.
Primary domain	Internal IT equipment and related inventory operations.
Primary users	Admin, IT Manager and Employee.
Primary interface	Responsive browser-based web application.
Core records	Assets, inventory items, vendors, purchases, employees, assignments, returns and maintenance.

2. Problem Statement

Organizations may rely on spreadsheets, email or disconnected records to manage IT equipment. This can make it difficult to keep stock quantities accurate, identify the current holder of an asset, trace purchase details, follow up on returns or understand maintenance status.

The proposed system will centralize these records and apply consistent business rules so that asset and inventory information remains traceable and up to date.

3. Objectives and Success Measures

3.1 Objectives

- Maintain a central record of company IT assets and stock-managed items.
- Record purchases and connect received items to inventory or individual asset records.
- Show available, assigned, under-maintenance, damaged and retired asset states.
- Record which employee currently holds each assigned asset.
- Track asset returns and inspection outcomes.
- Record maintenance issues, progress and completion.
- Provide useful operational summaries and searchable reports.

3.2 Success measures

The MVP will be considered useful when authorized users can complete the core workflows without relying on a separate spreadsheet as the primary record. Key measures include accurate stock balances, no simultaneous active assignment of one serialized asset, traceable assignment/return history, and complete purchase and maintenance records.

4. Users, Roles and Permissions

The first release supports three roles. Access must be enforced by the backend, not only by hiding frontend controls.

Role	Main responsibilities	Access expectation
Admin	Manage users and roles; access all modules; maintain master data; view all reports; perform administrative corrections.	Full system access, including user and configuration management.
IT Manager	Manage assets, inventory, vendors, purchases, employees, assignments, returns and maintenance; view operational reports.	Operational access to asset and inventory workflows. Cannot manage system-wide user roles unless explicitly granted.
Employee	View own assigned assets; submit an issue or maintenance request; view own request status and assignment details.	Limited access to own records and permitted request actions only.

4.1 Permission principles

- Users must authenticate before accessing protected application features.
- Each user must have one active role for the MVP unless a later requirement approves multiple roles.
- Employees must not view another employee's private assignment or request details.
- Only authorized roles can create, edit, approve or change operational records.
- Sensitive actions such as changing roles, retiring assets or correcting stock must be restricted and recorded.

4.2 Employee access decision

For this PRD, employees are included as authenticated users with limited access. They can view their own assigned assets and submit asset issue requests. They cannot manage inventory, purchases, vendors or assignments.

5. Scope and Boundaries

5.1 Included in the MVP

- Authentication and role-based access for Admin, IT Manager and Employee.
- Dashboard with key asset and inventory summaries.
- Asset categories and asset records, including unique identifiers for serialized equipment.
- Quantity-based inventory tracking for bulk accessories or stock-managed items.
- Vendor records and purchase records.
- Employee records.
- Asset assignment, return and return-condition inspection.
- Maintenance request and maintenance status tracking.
- Basic search, filters and operational reports.
- Audit history for important inventory and asset lifecycle actions.

5.2 Not included in the MVP

- Multiple-company or multi-tenant management.
- Native Android or iOS application.
- Barcode or QR-code scanning.
- Automated email, SMS or WhatsApp notifications.
- Accounting, payroll, procurement approval chains or ERP integrations.
- AI-based recommendations or predictive maintenance.

- Advanced depreciation, financial valuation or tax calculations.
- Complex purchase-order approval and vendor payment processing.

Items outside the MVP may be considered later only after the core workflows are stable.

6. Product Modules and Functional Requirements

Each requirement has an identifier so that it can be referenced in the development plan, implementation tasks and testing.

6.1 Authentication and User Management

ID	Requirement
AUTH-01	The system shall allow registered users to sign in using their credentials.
AUTH-02	The system shall allow users to sign out.
AUTH-03	The system shall restrict protected pages and APIs to authenticated users.
AUTH-04	The system shall enforce role-based permissions for Admin, IT Manager and Employee.
AUTH-05	Admin shall be able to create, update, activate and deactivate user accounts.
AUTH-06	The system shall store passwords using a secure password-hashing method and shall never store plain-text passwords.

6.2 Dashboard

ID	Requirement
DASH-01	Display total serialized assets and stock-managed item quantities as separate measures.
DASH-02	Display counts of available, assigned, under-maintenance, damaged and retired serialized assets.
DASH-03	Display recent purchases, assignments and maintenance activity.
DASH-04	Display pending maintenance requests requiring attention.
DASH-05	Apply role-based visibility so users see only information they are permitted to access.

6.3 Asset Management

Asset Management stores an individual record for each trackable piece of equipment, such as a laptop, desktop or monitor. Each serialized asset must have a unique asset identifier.

ID	Requirement
AST-01	Create an individual asset record with a unique asset ID.
AST-02	Store asset name, category, brand, model, serial number, purchase reference, purchase date, cost, warranty expiry, location and status where applicable.
AST-03	View asset details and lifecycle history.
AST-04	Edit permitted asset details while preserving important history.
AST-05	Search and filter assets by ID, name, category, serial number, status, employee and location where applicable.
AST-06	Prevent duplicate serial numbers where a serial number is supplied and is required to be unique.
AST-07	Change asset status only through permitted actions and business rules.

6.4 Inventory Management

Inventory Management tracks stock quantities for items managed in bulk, such as keyboards, mice, cables or adapters. Serialized assets are tracked individually and must not be counted twice as separate bulk stock.

ID	Requirement
INV-01	Create and maintain inventory item records with item name, category, unit of measure and reorder threshold where relevant.

ID	Requirement
INV-02	Show current quantity on hand for each stock-managed item.
INV-03	Record stock increases from confirmed purchase receipts.
INV-04	Record stock decreases for approved issue, consumption, adjustment or other defined stock movement.
INV-05	Maintain a dated stock movement history with movement type, quantity, reference and responsible user.
INV-06	Show low-stock status when quantity on hand reaches or falls below the configured threshold.
INV-07	Prevent stock from becoming negative unless a future approved business rule explicitly allows it.

6.5 Vendor Management

ID	Requirement
VEN-01	Create, view and update vendor records.
VEN-02	Store vendor name, contact person, email, phone number and address where available.
VEN-03	Search and filter vendor records.
VEN-04	Show purchase history associated with a vendor.
VEN-05	Prevent deletion of a vendor record that is referenced by existing purchase records; allow deactivation instead.

6.6 Purchase Management

ID	Requirement
PUR-01	Create a purchase record linked to a vendor.
PUR-02	Record purchase reference or invoice number, purchase date, received date, item, quantity, unit cost and notes where applicable.
PUR-03	Support multiple line items in one purchase record.
PUR-04	Record whether a purchase is draft, received or cancelled.
PUR-05	Update inventory only when items are confirmed as received, not merely when a draft purchase is created.
PUR-06	For serialized equipment, create or register one individual asset record per physical unit received.
PUR-07	For bulk items, increase the relevant stock-managed inventory quantity by the received quantity.
PUR-08	Prevent a cancelled or draft purchase from increasing available stock.
PUR-09	Show purchase history and linked vendor/item details.

6.7 Employee Management

ID	Requirement
EMP-01	Create, view and update employee records.
EMP-02	Store employee ID, name, department, designation, work email, contact details and employment status where available.
EMP-03	Search employees by ID, name or department.
EMP-04	Show each employee's currently assigned assets and assignment history to authorized users.
EMP-05	Prevent new assignments to inactive employees.

6.8 Asset Assignment and Return

ID	Requirement
ASN-01	Assign an available serialized asset to an active employee.
ASN-02	Record assignment date, assigned-by user, employee, asset and expected return date if applicable.
ASN-03	Prevent an asset from having more than one active assignment.
ASN-04	Update the asset status to Assigned when assignment is confirmed.
ASN-05	Record the actual return date and return condition.
ASN-06	After inspection, make a good-condition returned asset Available; route a damaged asset to Under Maintenance or Damaged as appropriate.
ASN-07	Retain assignment and return history; do not overwrite old assignment records.
ASN-08	Allow authorized users to view overdue expected returns.

6.9 Maintenance Management

ID	Requirement
MNT-01	Allow an employee to submit an issue request for an asset assigned to them.
MNT-02	Allow an IT Manager to create a maintenance record for an asset.
MNT-03	Record asset, issue description, reported date, reported-by user, priority if configured and status.
MNT-04	Support maintenance statuses: Reported, In Progress, Completed and Cancelled.
MNT-05	Record repair notes, service provider or responsible person, repair cost and completion date where applicable.
MNT-06	Keep maintenance history linked to the asset.
MNT-07	Prevent an asset under maintenance from being assigned until it is inspected and marked Available.
MNT-08	Allow employees to view the status of their own submitted issue requests.

6.10 Reports and Search

ID	Requirement
REP-01	Provide an asset register report with relevant filters.
REP-02	Provide an available and assigned asset report.
REP-03	Provide a stock quantity and low-stock report for bulk inventory.
REP-04	Provide purchase history by date range and vendor.
REP-05	Provide employee-wise asset assignment report.
REP-06	Provide maintenance status and maintenance history reports.
REP-07	Provide warranty-expiry information for assets with warranty dates.
REP-08	Restrict report data according to the user's role and permissions.

7. End-to-End Workflows

The following workflows describe the expected business process. The development plan should translate them into screens, APIs, database transactions and tests.

7.1 Purchase and receiving workflow

- Admin or IT Manager selects an existing vendor or creates a vendor record.
- Authorized user creates a purchase and adds one or more line items.
- Purchase remains Draft until goods are confirmed as received.
- On receipt, the user confirms the received quantity for each line item.
- For serialized equipment, the user enters the details of each physical unit and the system creates an individual asset record.
- For bulk items, the system adds the received quantity to the inventory balance and records a stock movement.
- The purchase and receiving details remain available for audit and reporting.

7.2 Asset assignment workflow

- IT Manager searches for an Available asset.
- IT Manager selects an active employee and the asset to assign.
- The system validates that the asset is Available and has no active assignment.
- The system records the assignment and changes the asset status to Assigned.
- The employee can view the assigned asset in their own account.

7.3 Asset return workflow

- IT Manager opens the employee's active assignment and records the return.
- The system records return date and condition notes.
- IT Manager inspects the returned equipment.
- If it is in good condition, the asset becomes Available.
- If it is damaged, the asset becomes Under Maintenance or Damaged according to the inspection outcome.
- The previous assignment remains in history.

7.4 Maintenance workflow

- Employee submits an issue for an asset assigned to them, or IT Manager creates a maintenance record.
- IT Manager reviews the issue and updates its status to In Progress when work starts.
- Repair details and costs are recorded where applicable.
- After completion, the asset is inspected.
- The asset becomes Available only after it is confirmed fit for use; otherwise it remains unavailable or is marked Damaged/Retired as appropriate.
- The request and maintenance history remain linked to the asset.

7.5 Stock issue workflow for bulk items

- Authorized user selects a stock-managed item and enters the quantity issued or consumed.
- The system validates that sufficient quantity is available.
- The system reduces the quantity on hand and records a stock movement with reason, reference and responsible user.
- The movement is visible in stock history and reports.

8. Business Rules and Validations

Rule ID	Business rule
BR-01	Every serialized asset must have a unique asset ID.
BR-02	A serialized asset can have only one active assignment at a time.
BR-03	Only Available assets can be assigned.
BR-04	A returned asset must be inspected before it becomes Available.
BR-05	An asset under maintenance, damaged or retired cannot be assigned.
BR-06	A draft or cancelled purchase must not increase inventory.
BR-07	Only confirmed received quantities update inventory.
BR-08	Serialized equipment creates individual asset records; bulk items update quantity-based stock.
BR-09	A stock movement must record item, quantity, movement type, date/time, reference/reason and responsible user.
BR-10	Stock quantity must not become negative.
BR-11	An inactive employee cannot receive a new asset assignment.
BR-12	Records referenced by transactions should be deactivated or archived rather than hard-deleted.
BR-13	Important changes to asset status, assignment, return, purchase receipt and stock must be traceable.
BR-14	All date/time values must be stored consistently and displayed in the organization's configured timezone.

8.1 Status definitions

Status	Meaning
Available	Asset is in usable condition and can be assigned.
Assigned	Asset is currently assigned to an employee.
Under Maintenance	Asset is being inspected, repaired or serviced and cannot be assigned.
Damaged	Asset is not fit for use and is awaiting a decision or repair.
Retired	Asset is permanently removed from active use and cannot be assigned.

9. Data Requirements

The following is a logical data inventory, not the final physical database schema. The development plan must propose exact columns, types, keys, indexes, constraints and relationships.

Entity	Purpose / key information
User	Login identity, password hash, role, account status and linked employee where applicable.
Role	Role name and permission definition.
Employee	Employee ID, name, department, designation, contact details and employment status.
Vendor	Vendor identity and contact information.
Asset Category	Category name and description for equipment classification.
Asset	Unique asset ID, serial number, model, category, purchase reference, warranty, location and current status.
Inventory Item	Bulk item identity, category, unit of measure, reorder threshold and current balance or derived balance.
Purchase	Vendor, purchase reference, dates, status, created-by user and notes.
Purchase Item	Purchase line item, item type, quantity, unit cost and received quantity.
Stock Movement	Inventory item, movement type, quantity, date/time, reason/reference and responsible user.
Asset Assignment	Asset, employee, assignment date, assigned-by user, expected return and active/closed state.
Asset Return	Assignment reference, return date, condition, inspection notes and outcome.
Maintenance Request	Asset, requester, issue description, status, priority if used, reported and completion dates.
Maintenance History	Status changes, work notes, service details, repair cost and responsible person.
Audit Log	Actor, action, affected record, timestamp and relevant before/after details for critical events.

9.1 Data relationship expectations

- One vendor can be linked to many purchases.
- One purchase can contain multiple purchase line items.
- One purchase line item can represent a serialized asset type or a bulk inventory item; the model must prevent ambiguous double-counting.
- One asset category can classify many assets or inventory items.
- One employee can have many historical assignments but should have only one active assignment per serialized asset.
- One asset can have many maintenance requests and maintenance history records.
- One bulk inventory item can have many stock movements.
- Each critical transaction must retain the user who performed it and its timestamp.

9.2 Inventory modelling requirement

The design must clearly separate individually tracked assets from quantity-based stock. For example, each laptop should have its own asset record and serial number. Keyboards or cables may be tracked by quantity. The same physical units must never be counted in both models.

The development plan should recommend whether current bulk stock is stored as a balance with transaction history or derived from stock movements, and explain how consistency will be maintained.

10. Non-Functional Requirements

Area	Requirement
Security	Use secure password hashing, authenticated sessions/tokens, backend-enforced authorization, input validation and protection against common web vulnerabilities.
Privacy	Employees may access only their own assignment and issue-request information; administrative access must be role-controlled.
Performance	Common list and detail pages should respond promptly under normal expected usage; use pagination and server-side filtering for growing datasets.
Reliability	Critical operations such as purchase receipt, assignment and stock issue must be transactional so partial updates do not leave inconsistent data.
Data integrity	Use database constraints for unique identifiers, valid references and quantity rules where appropriate.
Usability	Provide clear navigation, consistent forms, useful empty states, readable tables, confirmation for consequential actions and understandable error messages.
Compatibility	Support current mainstream desktop browsers and a responsive layout for tablet/mobile browser use.
Maintainability	Use a modular codebase, documented APIs, migrations, environment-based configuration and automated tests for core business rules.
Auditability	Record who performed important actions and when; preserve transaction history.
Backup and recovery	Define a database backup and restore approach before production deployment.

11. Technology Stack and System Architecture

The following stack is selected as the project baseline. The development plan may recommend a small adjustment only when it explains a clear technical reason and obtains approval before implementation.

Layer / Concern	Selected technology	Purpose
Frontend	React.js	Build the browser-based user interface and reusable components.
Language	JavaScript	Implement frontend interactions and API communication.
Backend	Python + FastAPI	Implement business logic and expose REST APIs.
Database	PostgreSQL	Persist structured business and transaction data.
ORM	SQLAlchemy	Map Python models to relational tables and manage database queries.
Validation	Pydantic	Validate request and response data.
Database migrations	Alembic	Version and apply database schema changes.
Authentication	JWT-based authentication	Carry authenticated identity for protected API requests; apply secure token handling.
API documentation	FastAPI OpenAPI / Swagger UI	Document and explore backend endpoints.
API testing	Postman	Manually test API requests and responses.
Version control	Git + GitHub	Track code changes and collaborate/manage source code.
Development environment	VS Code	Edit and run the application locally.

11.1 High-level architecture

Step	Component	Responsibility
1	React frontend	Collect user input, display data and call backend endpoints.
2	HTTP / REST API	Transport requests and responses between frontend and backend.
3	FastAPI backend	Authenticate requests, authorize roles, validate data and execute business rules.
4	SQLAlchemy	Perform structured database operations through Python models and queries.
5	PostgreSQL	Store users, assets, purchases, inventory, assignments, returns, maintenance and audit records.

11.2 Architecture principles

- Frontend must not connect directly to PostgreSQL; all data access goes through backend APIs.
- Business rules that affect stock, assignment and asset status must be enforced by the backend.
- Use database transactions for operations that update multiple related records.
- Keep configuration and secrets outside source code using environment variables.
- Separate development and production configuration.
- Use API schemas for validated request and response contracts.

12. API Requirements

The backend must expose documented REST APIs. Exact endpoint paths and payload schemas will be finalized in the technical design. The following API groups are required:

API group	Expected capabilities
Authentication	Login, current-user details and logout/token lifecycle as appropriate.
Users and roles	User creation, update, activation/deactivation and role administration.
Dashboard	Role-filtered summary counts and recent activity.
Assets	Create, list, detail, update, search/filter and permitted status actions.
Inventory	Item setup, current balances, stock movement history and authorized adjustments/issues.
Vendors	Create, list, detail, update, search and purchase history.
Purchases	Create purchase, manage line items, receive quantities, cancel where permitted and view history.
Employees	Create, list, detail, update, activate/deactivate and assignment history.
Assignments and returns	Assign asset, list active/history, record return and inspection outcome.
Maintenance	Create request, list/detail, update status and record completion details.
Reports	Filtered report data for assets, inventory, purchases, assignments and maintenance.

12.1 API standards

- Use appropriate HTTP methods and meaningful status codes.
- Validate request bodies and query parameters.
- Return consistent success and error response structures.
- Enforce authentication and role permissions on protected endpoints.
- Support pagination, sorting and filtering for list endpoints where needed.
- Document request fields, response fields, validation rules and common errors in OpenAPI/Swagger.
- Do not expose passwords, password hashes, secrets or unauthorized employee information.

13. UI/UX Requirements

- Provide a clean, professional and consistent administrative interface.
- Use clear navigation for Dashboard, Assets, Inventory, Vendors, Purchases, Employees, Assignments, Maintenance and Reports.
- Show role-appropriate navigation and actions.
- Use searchable and filterable tables for records with clear status labels.
- Use forms with field labels, required-field indicators and helpful validation messages.
- Show confirmation before consequential actions such as receiving purchases, assigning assets, returning assets, retiring assets or adjusting stock.
- Provide loading, empty, success and error states.
- Make important record details and history easy to find.
- Use responsive layouts suitable for desktop-first internal operations and smaller browser screens.

13.1 Main screens expected

Screen	Main content
Login	Credential form and validation messages.
Dashboard	Summary cards, recent activity and pending items.
Asset list and detail	Search/filter list, asset information, status and history.
Inventory list and detail	Item balances, reorder threshold and stock movement history.
Vendor list and detail	Vendor contact information and purchase history.
Purchase list and form	Vendor, line items, quantities, costs, receipt status and history.
Employee list and detail	Employee information and assigned asset history.
Assignment and return	Select employee/asset, assignment information, return and inspection actions.
Maintenance	Issue requests, status, repair information and history.
Reports	Report filters, summary and tabular results.

14. Reporting Requirements

The MVP reports should help users answer practical operational questions. Reports must use the same source records as the operational modules.

Report	Key information / filters
Asset Register	Asset ID, name, category, serial number, status, location and assigned employee; filter by category/status/location.
Asset Availability	Available, assigned, maintenance, damaged and retired counts; filter by category.
Inventory Stock	Bulk item, quantity on hand, reorder threshold and low-stock flag.
Purchase History	Vendor, purchase reference, date, item, ordered/received quantity and status; filter by date/vendor.
Employee Asset List	Employee and their current/historical assigned assets.
Maintenance Report	Asset, issue, status, reported date, completion date and repair cost where recorded.
Warranty Expiry	Assets with warranty dates within a selected date range.

15. Acceptance Criteria

The MVP is ready for stakeholder review when the following end-to-end checks pass:

- An authorized Admin can sign in, create users and assign the supported roles.
- An IT Manager can create a vendor and record a purchase with multiple line items.
- Draft or cancelled purchases do not change stock; confirmed receipts update the correct inventory model.
- Serialized equipment received through a purchase can be registered as individual assets with unique identifiers.
- Bulk items update quantity-based stock and create traceable stock movements.
- An available serialized asset can be assigned to an active employee and becomes Assigned.
- The same asset cannot be assigned to a second employee while an active assignment exists.
- A returned asset is inspected and becomes Available or unavailable according to its condition.
- An employee can view their own assigned assets and submit an issue request, but cannot access another employee's records.
- Maintenance status and history can be updated by an authorized IT Manager.
- An asset under maintenance, damaged or retired cannot be assigned.
- Reports show data consistent with the underlying records and respect role permissions.
- Validation errors are understandable and invalid operations do not corrupt stock or asset records.
- Critical actions retain the responsible user and timestamp.

16. MVP Delivery Priorities

The following sequence is a recommended dependency order, not a commitment to a fixed timeline. The detailed development plan should estimate effort after reviewing the requirements.

Phase	Focus	Expected result
1. Foundation	Project setup, configuration, database connection, migrations, authentication and roles.	Users can sign in and access authorized areas.
2. Master data	Employees, vendors, asset categories and inventory item setup.	Core reference records are available.
3. Purchase and receiving	Purchase records, line items, receipt and stock/asset creation.	Purchases reliably create inventory or asset records.
4. Asset and inventory operations	Asset register, stock balances, movements, search and filters.	Users can view and manage assets and quantities.
5. Assignment and return	Employee assignment, return, inspection and history.	Asset custody is traceable.
6. Maintenance	Issue requests, maintenance status and history.	Faults and repairs can be tracked.
7. Dashboard and reports	Summary views and operational reports.	Users can monitor operations.
8. Quality and release readiness	Integration tests, security checks, usability review, backup and deployment preparation.	MVP is ready for stakeholder acceptance.

16.1 Suggested testing levels

- Unit tests for business rules such as stock updates, assignment restrictions and status transitions.
- API tests for authentication, authorization, validation and response behavior.
- Integration tests for purchase receipt, assignment, return and maintenance workflows.
- UI tests or manual acceptance checks for critical user journeys.
- Regression tests after changes to inventory or asset lifecycle logic.

17. Risks, Assumptions and Open Decisions

17.1 Assumptions used in this PRD

- The system is for one company, not a multi-tenant SaaS product.
- The initial interface is a web application.
- Employees have limited login access to view their own assets and submit issue requests.
- IT Manager performs operational asset, purchase, inventory and maintenance activities.
- Asset purchases may include serialized equipment and bulk stock-managed items.
- Only received quantities update inventory.
- The organization will provide employee, vendor and initial asset data for setup or data entry.

17.2 Decisions to confirm during detailed planning

Decision	Current baseline / question
Locations	Support a basic location field for assets. Confirm whether multiple offices, rooms or storage locations need structured management.
Purchase receiving	MVP assumes a single receiving confirmation per purchase line. Confirm whether partial receipts are required.
Stock adjustments	Allow authorized adjustments with a mandatory reason and audit record. Confirm whether approval is required.
Maintenance ownership	IT Manager manages requests. Confirm whether a separate technician role is needed later.
Warranty reminders	MVP includes warranty-expiry reporting; automated notifications are future scope.
Exports	Confirm whether CSV/PDF export is required for the first release or can be postponed.
Deployment	Hosting provider, domain, backup frequency and production environment are to be decided before release.

Open decisions must be raised for stakeholder confirmation. The development team must not silently invent business rules for unresolved items.

18. Glossary

Term	Meaning
IT Asset	An individually identifiable piece of company equipment, such as a laptop or monitor.
Serialized Asset	An asset tracked individually using a unique asset ID and/or serial number.
Inventory Item	A product tracked by quantity, such as keyboards, mice, cables or adapters.
Stock on Hand	The quantity currently available in inventory records.
Stock Movement	A recorded increase or decrease in a stock-managed item's quantity.
Assignment	A record showing that a serialized asset has been issued to an employee.
Return Inspection	A check of the asset's condition when it is returned.
Maintenance Request	A record of a reported issue requiring inspection, repair or service.
MVP	Minimum Viable Product: the first usable release containing the agreed essential features.

19. Instructions for Claude: Development Plan

Use this PRD as the source of truth. Analyze it and prepare a detailed, beginner-understandable development plan. Do not generate the full application code yet.

19.1 Required output

- Summarize the product, its users and the main workflows to confirm understanding.
- Identify ambiguities, conflicts, missing requirements and decisions that need stakeholder confirmation.
- Propose the system architecture using the selected stack: React.js, JavaScript, Python, FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Alembic and JWT.
- Explain the responsibility of each technology in simple terms and show how a frontend request reaches the database and returns a response.
- Propose a frontend folder structure and explain the purpose of the main folders and files.
- Propose a backend folder structure organized by modules, schemas, models, routes, services and configuration.
- Design the relational database: tables, columns, data types, primary keys, foreign keys, constraints, indexes and relationships.
- Explain the serialized-asset versus quantity-based-inventory model and how duplicate counting will be prevented.
- List REST API endpoints by module, including HTTP method, purpose, role access, request fields, response shape and validation/error cases.
- Explain authentication and role-based authorization, including what each role can and cannot do.
- Describe the key business logic and database transactions for purchase receipt, stock movement, assignment, return and maintenance.
- Provide a screen-by-screen frontend plan and identify reusable UI components.
- Map each PRD requirement ID to the planned module, API, data entities and tests.
- Provide a dependency-aware implementation sequence with small milestones and a definition of done for each milestone.
- Provide a testing plan covering unit, API, integration and user acceptance testing.
- Identify security, data integrity, backup and deployment considerations.

19.2 Planning rules

- Do not add features outside the MVP without clearly labelling them as suggestions for later.
- Do not change the selected technology stack without explaining why and requesting approval.
- Do not assume unresolved business decisions; list them as questions.
- Do not duplicate the same business responsibility across multiple modules.
- Keep the plan practical for a beginner developer learning while building.
- Explain technical terms when they first appear.
- Prioritize a working end-to-end workflow over building every screen independently.
- Do not write the entire application code until the development plan has been reviewed and approved.

19.3 Expected plan format

Deliverable	Expected detail
A. Understanding	Product summary, user roles and workflow summary.
B. Questions	Missing or ambiguous requirements that need decisions.
C. Architecture	Component diagram and technology responsibilities.
D. Data design	ER diagram description, schema and relationship explanations.
E. API design	Endpoint catalogue, payloads, access and error handling.
F. UI plan	Pages, navigation, forms, tables and reusable components.
G. Implementation roadmap	Phases, dependencies, tasks and completion checks.
H. Quality plan	Tests, security, data integrity and release checklist.

Final instruction
Before coding, present the full development plan and list all questions that require approval. Wait for the project owner to review and approve the plan. This PRD remains the business requirements baseline.

End of Product Requirements Document